

Homework 4 - due Friday February 20

1. Let R be a UFD, and let a and b be nonzero non-unit elements of R . Prove that a and b have a gcd, and describe it in terms of their prime factorizations.
2. Let R be a PID and let S be a multiplicative set of R , and let $P \subseteq R$ be a prime ideal.
 - (a) Prove that R/P is a PID
 - (b) Prove that $S^{-1}R$ is a PID.
3. Let R be a Noetherian UFD.
 - (a) Prove that every descending chain of prime ideals $P_0 \supseteq P_1 \supseteq \cdots \supseteq P_n \supseteq \cdots$ in R stabilizes.
 - (b) The **height** of a prime ideal P is (if finite) the maximum length h of a chain of prime ideals $P_0 \subseteq P_1 \subseteq \cdots \subseteq P_h = P$. Prove that every prime ideal in R of height 1 is principal.
4. Consider the ring $R := \mathbb{Z}[\sqrt{-5}] := \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$.
 - (a) Prove that $R \cong \mathbb{Z}[x]/(x^2 + 5)$.
 - (b) Since R is a subring of $\mathbb{Q}(\sqrt{d})$ where $d = -5$, the norm N defined in Homework 2 will still have the property that $N(zw) = N(z)N(w)$ in R . Use this to prove that the units in R are ± 1 .
 - (c) Prove that $2, 3, (1 + \sqrt{-5})$ and $(1 - \sqrt{-5})$ are all irreducible non-associate elements of R .
 - (d) Prove that R is not a UFD.
5. Let R be a Euclidean domain which is not a field. Prove that there exists a nonzero, non-unit element $c \in R$ such that for all $a \in R$, there is some $q, r \in R$ such that $a = qc + r$ and $r = 0$ or r is a unit. (Hint: Pick c so that $N(c)$ is a minimal positive value among norms of non-units).
6. Consider the ring $R := \mathbb{Z}[\frac{1+\sqrt{-19}}{2}] := \{a + b\frac{1+\sqrt{-19}}{2} \mid a, b \in \mathbb{Z}\}$.
 - (a) Since R is a subring of $\mathbb{Q}(\sqrt{d})$ where $d = -19$, the norm N defined in Homework 2 will still have the property that $N(zw) = N(z)N(w)$ in R . Prove that the smallest values of $N(z)$ for $z \in R$ are $0, 1, 4, 5$ and that $N(a + b\frac{1+\sqrt{-19}}{2}) \geq 5$ if $b \neq 0$.
 - (b) Prove that the units in R are ± 1 .
 - (c) Prove that if $c \in R$ satisfies the condition in the previous exercise, then c must divide 2 or 3 in R , and conclude $c = \pm 2$ or ± 3 .
 - (d) Show that there is no q, r such that $\frac{1+\sqrt{-19}}{2} = qc + r$ with $c = \pm 2, \pm 3$ and $r = 0, \pm 1$, and conclude that R is not a Euclidean domain.